Agile/Lean

1. What is a "minimum unit of work" for your project? Think of this as a task breakdown to something that is actionable and assignable to a team member. This task breakdown is analogous to the work breakdown technique in Waterfall, or issue creation in Github, or filing a ticket in a ticketing system.

   Ans: For this project, a minimum unit of work is a small, actionable task that creates a measurable change in the analysis or communication of results. Each unit is self-contained, testable, and contributes directly to project progress. Examples include cleaning one variable, running a single statistical test, or building a chart tied to a specific research question. In practice, this could mean performing a “log transformation and validation of income distribution,” “running a Kruskal–Wallis test on payment delays,” or “publishing a Tableau visualization comparing utilization across score bands.” Agile encourages breaking work into the smallest deliverable steps and providing visible value through short, iterative cycles. Keeping units of work small helps identify issues early, gather timely feedback, and maintain concise documentation that is current but not burdensome. Each completed unit serves as a clear, reviewable outcome that moves the project forward while keeping it adaptable.

2. Evaluate the utility of Agile/Lean for your project. Is success just a matter of implementing Agile/Lean optimally, or is it fundamentally unsuitable for your project? Present your answer as a set of pros and a set of cons.

   Ans: Agile and Lean methodologies are well-suited for this project because the workflow naturally benefits from short iterations, regular feedback, and the ability to showcase intermediate results like updated models or new visualizations. The iterative approach allows quick adaptation when data quality issues or assumption violations arise, while Lean principles help eliminate unnecessary effort by focusing only on documentation and outputs that provide real value. These methods promote steady progress, clearer communication, and consistent improvement through visible, working deliverables. That said, success depends on applying Agile thoughtfully, not just mechanically. The project could face challenges such as an unclear backlog, weak prioritization, or dependencies on large datasets that don’t align neatly with short sprint cycles. When the backlog lacks focus or when data constraints slow progress, the flow can become inefficient. Therefore, while Agile and Lean principles fit the exploratory and iterative nature of this work, maintaining a clear vision, well-defined tasks, and meaningful reviews is essential for sustained success.

3. What kinds of technology are useful in enabling Agile teams? Consider the types of techniques and tools necessary for enabling the communication-first and distributed nature of Agile/Lean.

   Ans: Technology plays a key role in supporting Agile and Lean processes, especially when communication and visibility are central. Tools like GitHub, Jira, or Trello enable issue tracking, version control, and backlog management, transforming project goals into actionable items. For data analysis, tools such as Jupyter Notebooks, Google Colab, and Tableau make it easier to iterate on analyses and share real, working outputs with minimal friction. These technologies allow quick demos, reproducibility, and transparent progress tracking, reinforcing Agile's "working product over documentation" mindset. Communication-focused tools such as Slack or Microsoft Teams help resolve blockers and share updates efficiently, while lightweight documentation through README files, short guides, and "cookbook-style" snippets keeps knowledge current without creating unnecessary overhead. Light automation, such as running code checks or notebook tests, further supports continuous improvement. Together, these technologies make it easier to collaborate, share progress, and adapt quickly core strengths of Agile and Lean methods.

4. What are the components of an Agile Sprint? Consider each of these components in the context of your hypothetic.
   Ans: An Agile sprint for this project begins with sprint planning, where a short, specific goal is defined—such as validating a model assumption, improving data cleaning scripts, or creating a new dashboard visualization. After identifying the most important backlog items, the implementation phase focuses on completing those tasks and producing working outputs that can be reviewed. Throughout the sprint, brief daily reflections act as standups, helping identify obstacles and keep progress visible. The sprint concludes with a review, where completed deliverables such as updated feature importance charts or refined visualizations are demonstrated and evaluated for feedback. Finally, a retrospective helps reflect on what went well and what can be improved, such as tightening the definition of done or refining data validation steps. This continuous cycle of

planning, building, reviewing, and reflecting keeps the workflow adaptive and focused on delivering real value each iteration.